Chapter 3. Convergence of Fourier Series

Mean-square convergence and pointwise behaviour

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3.1 Mean-square convergence of Fourier series

Motivation. Pointwise convergence is delicate, but in the mean-square sense the Fourier series of every integrable function converges. The reason is geometric: $\{e^{in\theta}\}$ is an orthonormal family, $S_N(f)$ is the orthogonal projection onto trigonometric polynomials of degree $\le N$, and trigonometric polynomials are dense (Fejér).

Theorem 1.1 (Mean-square convergence)

If f is integrable on the circle, then

$$\frac{1}{2\pi}\int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2\, d\theta \to 0 \quad \text{as } N \to \infty.$$

3.1.1 Vector spaces and inner products

An **inner product** $(\cdot,\cdot)$ on a complex vector space V is linear in the first variable, $(Y,X) = \overline{(X,Y)}$, and $(X,X) \ge 0$; it is **strictly positive definite** if $(X,X) = 0$ only for $X = 0$. $\|X\| = (X,X)^{1/2}$. Basic facts, for any inner product:

- **Pythagoras:** if $X \perp Y$ ($(X,Y) = 0$), then $\|X+Y\|^2 = \|X\|^2 + \|Y\|^2$.
- **Cauchy–Schwarz:** $|(X,Y)| \le \|X\|\|Y\|$.
- **Triangle inequality:** $\|X+Y\| \le \|X\| + \|Y\|$.

Proofs [Supplement]. Pythagoras: expand. Cauchy–Schwarz: if $\|Y\| = 0$, then $0 \le \|X - tY\|^2 = \|X\|^2 - 2t\,\mathrm{Re}(X,Y)$ for all real t forces $\mathrm{Re}(X,Y) = 0$, and similarly $\mathrm{Im}(X,Y) = 0$. Otherwise, with $c = (X,Y)/\|Y\|^2$, $X - cY \perp Y$ and $\|X\|^2 = \|X - cY\|^2 + |c|^2\|Y\|^2 \ge |(X,Y)|^2/\|Y\|^2$. Triangle: $\|X+Y\|^2 = \|X\|^2 + 2\,\mathrm{Re}(X,Y) + \|Y\|^2 \le (\|X\| + \|Y\|)^2$.

Examples. (1) $\mathbb{R}^d$, $\mathbb{C}^d$ with the usual inner products.

(2) $\ell^2(\mathbb{Z})$: complex sequences $(a_n)_{n\in\mathbb{Z}}$ with $\sum |a_n|^2 < \infty$ and $(A,B) = \sum a_n \bar{b}_n$ (finite by Cauchy–Schwarz in $\mathbb{C}^d$ and a limit). It is a vector space (by $|a+b|^2 \le 2|a|^2 + 2|b|^2$) and **complete**: Cauchy sequences converge.

(3) $\mathcal{R}$: integrable functions on $[0,2\pi]$ with $(f,g) = \frac{1}{2\pi}\int_0^{2\pi} f\bar{g}$. This inner product is not strictly positive definite ($f = 0$ except at one point has $\|f\| = 0$), and $\mathcal{R}$ is not complete. Completing it leads to Lebesgue's L^2.

3.1.2 Proof of mean-square convergence

Let $e_n(\theta) = e^{in\theta}$; then $\{e_n\}$ is orthonormal in $\mathcal{R}$: $(e_n, e_m) = \frac{1}{2\pi}\int e^{i(n-m)\theta} d\theta = \delta_{nm}$. With $a_n = \hat{f}(n) = (f, e_n)$, $S_N(f) = \sum_{|n|\le N} a_n e_n$.

Key orthogonality. $f - S_N(f) \perp e_n$ for $|n| \le N$, since $(f - S_N(f), e_n) = a_n - a_n = 0$. Hence $f - S_N(f) \perp \sum_{|n|\le N} b_n e_n$ for all b_n, and in particular (Pythagoras)

$$\|f\|^2 = \|f - S_N(f)\|^2 + \sum_{|n|\le N} |a_n|^2. \tag{3.1}$$

Lemma 1.2 (Best approximation)

If f is integrable on the circle with Fourier coefficients a_n, then $\|f - S_N(f)\| \le \|f - \sum_{|n|\le N} c_n e_n\|$ for all complex c_n, with equality iff $c_n = a_n$ for all $|n| \le N$.

Proof. $f - \sum c_n e_n = (f - S_N(f)) + \sum b_n e_n$ with $b_n = a_n - c_n$. The two summands are orthogonal, so $\|f - \sum c_n e_n\|^2 = \|f - S_N(f)\|^2 + \sum |b_n|^2$. ∎

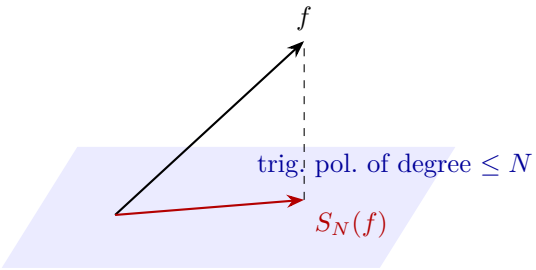


Figure. *$S_N(f)$ is the orthogonal projection of f, hence the best approximation (Lemma 1.2).*

Proof of Theorem 1.1.

Step 1 (continuous f). Given $\epsilon > 0$, Corollary 5.4 of Chapter 2 gives a trigonometric polynomial P of some degree M with $|f - P| < \epsilon$ everywhere, so $\|f - P\| < \epsilon$. By Lemma 1.2, $\|f - S_N(f)\| \le \|f - P\| < \epsilon$ for all $N \ge M$.

Step 2 (integrable f). Let $|f| \le B$. By Lemma 3.2 of Chapter 2 there is a continuous g with $|g| \le B$ and $\int |f - g| < \epsilon^2$, so $\|f - g\|^2 = \frac{1}{2\pi}\int |f-g|^2 \le \frac{2B}{2\pi}\int |f-g| < C\epsilon^2$. Choose a trigonometric polynomial P with $\|g - P\| < \epsilon$ (Step 1). Then $\|f - P\| < C'\epsilon$, and again Lemma 1.2 gives $\|f - S_N(f)\| < C'\epsilon$ for N large. ∎

Theorem 1.3 (Parseval)

Let $f \sim \sum a_n e^{in\theta}$ be integrable on the circle. Then:
(i) Mean-square convergence: $\frac{1}{2\pi}\int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2 d\theta \to 0$.
(ii) Parseval's identity: $\sum_{n=-\infty}^{\infty} |a_n|^2 = \frac{1}{2\pi}\int_0^{2\pi} |f(\theta)|^2 d\theta$.
(iii) The map $f \mapsto (a_n)_{n\in\mathbb{Z}}$ is "unitary": $(f, g) = \sum a_n \bar{b}_n$ for $g \sim \sum b_n e^{in\theta}$ (Lemma 1.5).

Proof. (i) is Theorem 1.1.

(ii) Let $N \to \infty$ in (3.1).

(iii) is Lemma 1.5 below. ∎

Bessel's inequality $\sum |a_n|^2 \le \|f\|^2$ already follows from (3.1).

Theorem 1.4 (Riemann–Lebesgue lemma)

If f is integrable on the circle, then $\hat{f}(n) \to 0$ as $|n| \to \infty$. Equivalently, $\int_0^{2\pi} f(\theta) \sin N\theta \, d\theta \to 0$ and $\int_0^{2\pi} f(\theta) \cos N\theta \, d\theta \to 0$ as $N \to \infty$.

Proof. By Parseval $\sum |\hat{f}(n)|^2 < \infty$, so the terms tend to 0. The reformulation follows from $\cos N\theta = \frac{e^{iN\theta} + e^{-iN\theta}}{2}$, $\sin N\theta = \frac{e^{iN\theta} - e^{-iN\theta}}{2i}$. ∎

Lemma 1.5

If $F \sim \sum a_n e^{in\theta}$ and $G \sim \sum b_n e^{in\theta}$ are integrable on the circle, then $\frac{1}{2\pi}\int_0^{2\pi} F(\theta)\overline{G(\theta)}\,d\theta = \sum_{n=-\infty}^{\infty} a_n \bar{b}_n$, the series converging absolutely.

Proof. [Supplement] Absolute convergence: $|a_n\bar{b}_n| \le \frac{1}{2}(|a_n|^2 + |b_n|^2)$ and Parseval. Polarization: $(F,G) = \frac{1}{4}\sum_{k=0}^{3} i^k \|F + i^k G\|^2$ and the same identity in $\ell^2(\mathbb{Z})$, i.e. $\sum a_n\bar{b}_n = \frac{1}{4}\sum_k i^k \sum_n |a_n + i^k b_n|^2$; the Fourier coefficients of $F + i^k G$ are $a_n + i^k b_n$, and Parseval applies to each term. ∎

3.2 Return to pointwise convergence

Motivation. Differentiability at a point suffices for convergence there, and convergence depends only on the behaviour of f near the point. Mere continuity does not suffice: breaking the symmetry between $e^{in\theta}$ and $e^{-in\theta}$ produces a continuous function whose Fourier series diverges at 0.

3.2.1 A local result

Theorem 2.1

If f is integrable on the circle and differentiable at θ_0, then $S_N(f)(\theta_0) \to f(\theta_0)$.

Proof. Define, for $|t| \le \pi$,

$$F(t) = \begin{cases} \dfrac{f(\theta_0 - t) - f(\theta_0)}{t} & t \neq 0, \\ -f'(\theta_0) & t = 0. \end{cases}$$

F is bounded near 0 (differentiability) and integrable on $[-\pi, \pi]$ (Appendix, Proposition 1.4). Since $\frac{1}{2\pi}\int D_N = 1$,

$$S_N(f)(\theta_0) - f(\theta_0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} (f(\theta_0 - t) - f(\theta_0)) D_N(t)\,dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(t)\, t\, D_N(t)\,dt.$$

With $D_N(t) = \frac{\sin((N+1/2)t)}{\sin(t/2)}$ and $\sin((N+\frac{1}{2})t) = \sin Nt \cos\frac{t}{2} + \cos Nt \sin\frac{t}{2}$,

$$S_N(f)(\theta_0) - f(\theta_0) = \frac{1}{2\pi}\int F(t)\frac{t}{\sin(t/2)}\cos\tfrac{t}{2}\,\sin Nt\,dt + \frac{1}{2\pi}\int F(t)\,t\cos Nt\,dt.$$

The functions $F(t)\frac{t\cos(t/2)}{\sin(t/2)}$ and $tF(t)$ are integrable ($t/\sin(t/2)$ is bounded on $[-\pi,\pi]$), so both integrals tend to 0 by the Riemann–Lebesgue lemma (Theorem 1.4). ∎

(The same proof works if f is only Lipschitz at θ_0: $|f(\theta) - f(\theta_0)| \le M|\theta - \theta_0|$.)

Theorem 2.2 (Riemann localization principle)

If f, g are integrable on the circle and $f = g$ on an open interval I containing θ_0, then $S_N(f)(\theta_0) - S_N(g)(\theta_0) \to 0$.

Proof. $f - g$ vanishes in I, so it is differentiable at θ_0 with value 0 there; apply Theorem 2.1 to $f - g$. ∎

3.2.2 A continuous function with diverging Fourier series

Symmetry breaking. For the sawtooth $f(\theta) = i(\pi - \theta)$ on $0 < \theta < \pi$, odd, we have $f \sim \sum_{n \neq 0} \frac{e^{in\theta}}{n}$. The "half" $\sum_{n \leq -1} \frac{e^{in\theta}}{n}$ is not the Fourier series of any integrable (in particular bounded) $\tilde{f}$: otherwise $|A_r(\tilde{f})(0)| = \sum_{n \geq 1} \frac{r^n}{n} \to \infty$ as $r \to 1$, whereas $|A_r(\tilde{f})(0)| \leq \frac{1}{2\pi} \int |\tilde{f}| P_r \leq \sup |\tilde{f}|$. For $N \geq 1$ let

$$f_N(\theta) = \sum_{1 \leq |n| \leq N} \frac{e^{in\theta}}{n}, \qquad \tilde{f}_N(\theta) = \sum_{-N \leq n \leq -1} \frac{e^{in\theta}}{n}.$$

Then (i) $|\tilde{f}_N(0)| = \sum_{n=1}^{N} \frac{1}{n} \geq \log N$ (compare with $\int_1^N dx/x$), and (ii) $f_N(\theta)$ is bounded uniformly in N and θ, by the following lemma.

Lemma 2.3

Suppose the Abel means $A_r = \sum_{n \geq 1} r^n c_n$ of $\sum c_n$ are bounded as $r \to 1^-$, and $c_n = O(1/n)$. Then the partial sums $S_N = \sum_{n=1}^{N} c_n$ are bounded.

Proof. Let $r = 1 - 1/N$ and $n|c_n| \leq M$. Then

$$|S_N - A_r| \leq \sum_{n=1}^{N} |c_n|(1-r^n) + \sum_{n>N} r^n |c_n| \leq M \sum_{n=1}^{N} (1-r) + \frac{M}{N} \sum_{n>N} r^n \leq MN(1-r) + \frac{M}{N} \frac{1}{1-r} = 2M,$$

using $1 - r^n = (1-r)(1 + r + \cdots + r^{n-1}) \leq n(1-r)$. So if also $|A_r| \leq M$, then $|S_N| \leq 3M$. ∎

Apply it to $\sum_{n \geq 1} c_n$ with $c_n = \frac{e^{in\theta}}{n} - \frac{e^{-in\theta}}{n}$, whose partial sums are $f_N(\theta)$: $|c_n| \leq 2/n$, and the Abel means are $(f * P_r)(\theta)$, bounded by $\sup |f|$ since P_r is a good kernel. Hence $|f_N(\theta)| \leq C$ for all N, θ.

Now shift frequencies by $2N$: $P_N(\theta) = e^{i2N\theta} f_N(\theta)$, $\tilde{P}_N(\theta) = e^{i2N\theta} \tilde{f}_N(\theta)$, trigonometric polynomials whose nonzero coefficients sit at $N \leq n \leq 3N$, $n \neq 2N$ (resp. $N \leq n \leq 2N - 1$).

Lemma 2.4

$S_M(P_N) = P_N$ if $M \geq 3N$; $S_M(P_N) = \tilde{P}_N$ if $M = 2N$; $S_M(P_N) = 0$ if $M < N$.

Proof. S_M keeps exactly the frequencies $|n| \leq M$. The frequencies of P_N lie in $[N, 3N]$; those $\leq 2N$ are exactly the frequencies of $\tilde{P}_N$. ∎

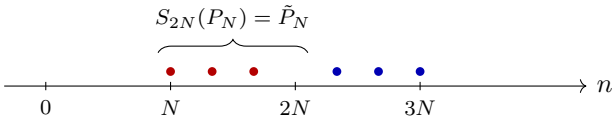


Figure. *Frequencies of P_N (centre $2N$); S_{2N} keeps only the left half and breaks the symmetry.*

The construction. Choose $\alpha_k > 0$ with $\sum \alpha_k < \infty$ and integers N_k with $N_{k+1} > 3N_k$ and $\alpha_k \log N_k \to \infty$; e.g. $\alpha_k = 1/k^2$, $N_k = 3^{2^k}$ (then $\alpha_k \log N_k = \frac{2^k \log 3}{k^2} \to \infty$). Let

$$f(\theta) = \sum_{k=1}^{\infty} \alpha_k P_{N_k}(\theta).$$

Since $|P_N(\theta)| = |f_N(\theta)| \leq C$, the series converges uniformly, so f is continuous, and its Fourier coefficients are obtained termwise. By Lemma 2.4 (the frequency ranges $[N_k, 3N_k]$ are disjoint and $3N_k < N_{k+1}$, so $2N_m < N_k$ for $k > m$ and $2N_m \geq 3N_k$ for $k < m$),

$$S_{2N_m}(f) = \sum_{k<m} \alpha_k P_{N_k} + \alpha_m \tilde{P}_{N_m}.$$

At $\theta = 0$: $|\sum_{k<m} \alpha_k P_{N_k}(0)| \le C \sum \alpha_k$, while $|\tilde{P}_{N_m}(0)| = |\tilde{f}_{N_m}(0)| \ge \log N_m$. So $|S_{2N_m}(f)(0)| \ge \alpha_m \log N_m - C' \to \infty$: *the Fourier series of the continuous function f diverges at* 0.